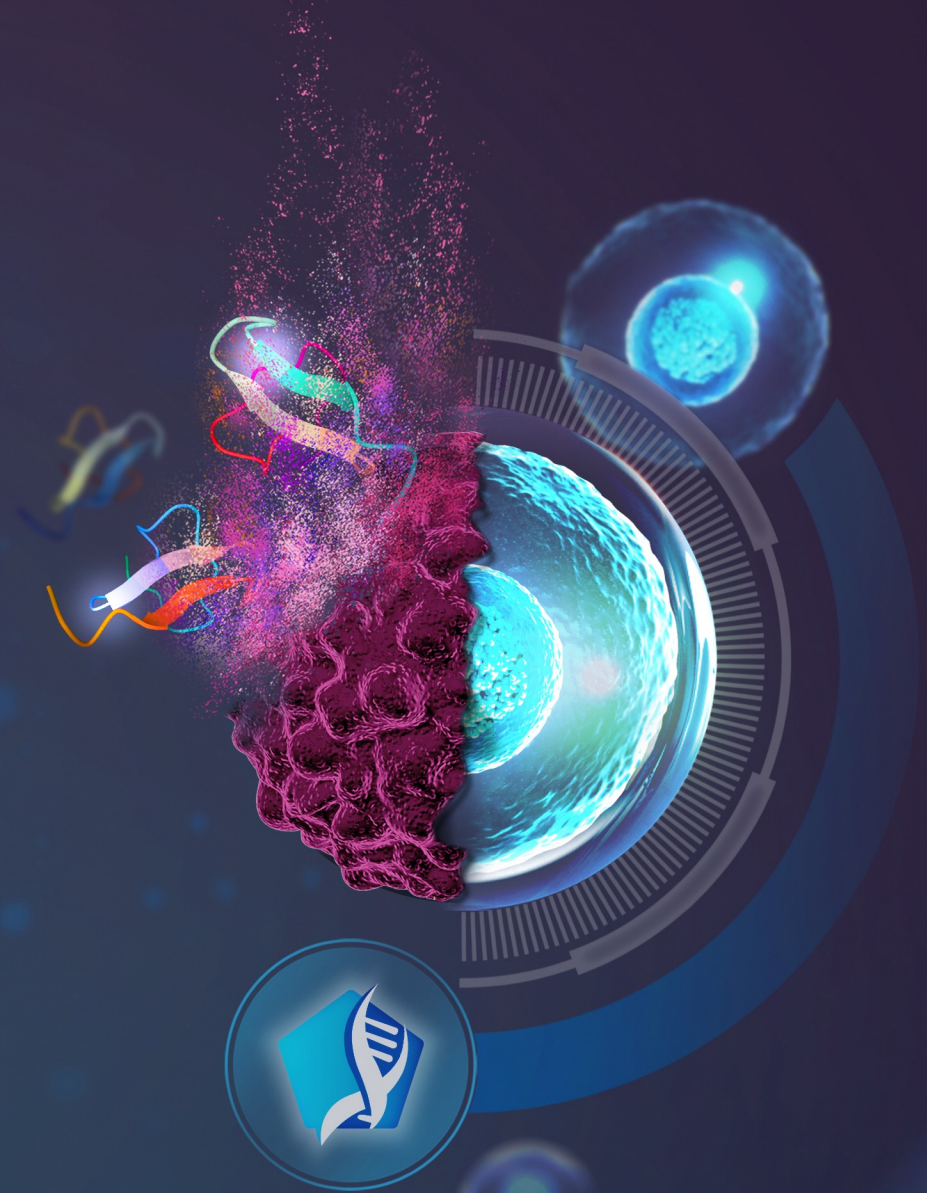


AI-BASED CANCER VACCINE PIONEER

Neogenlogic

Corporate Profile

Sep 2026



Neogenlogic at a Glance

A subsidiary of SCL Science, founded on a decade of neoantigen research at KAIST

Platform

DeepNeo · LexiNeo · DarkNeo

Proprietary AI, patents granted or filed

MHC presentation, TCR and BCR recognition,
and non-canonical translation

Pipeline

Four programs

Three indications, IND targeted for 2027

Three individualized and one off-the-shelf,
all mRNA-LNP

Markets

Korea and Japan first

MFDS and PMDA in parallel

Clinical sites engaged, CRO and
manufacturing partners secured

What we do

We design cancer vaccines by predicting which tumor antigens a patient's immune system will actually recognize.

Manufacturing is procured from established partners; the value we hold is the accuracy of that prediction and the intellectual property behind it.

The Modality Has Been Validated

Moderna, INTerpath-001, 2026. The first Phase 3 success for an individualized neoantigen therapy

1,137 patients

Randomized controlled Phase 3

RFS and DMFS

Primary and key secondary endpoints met

49% / 59%

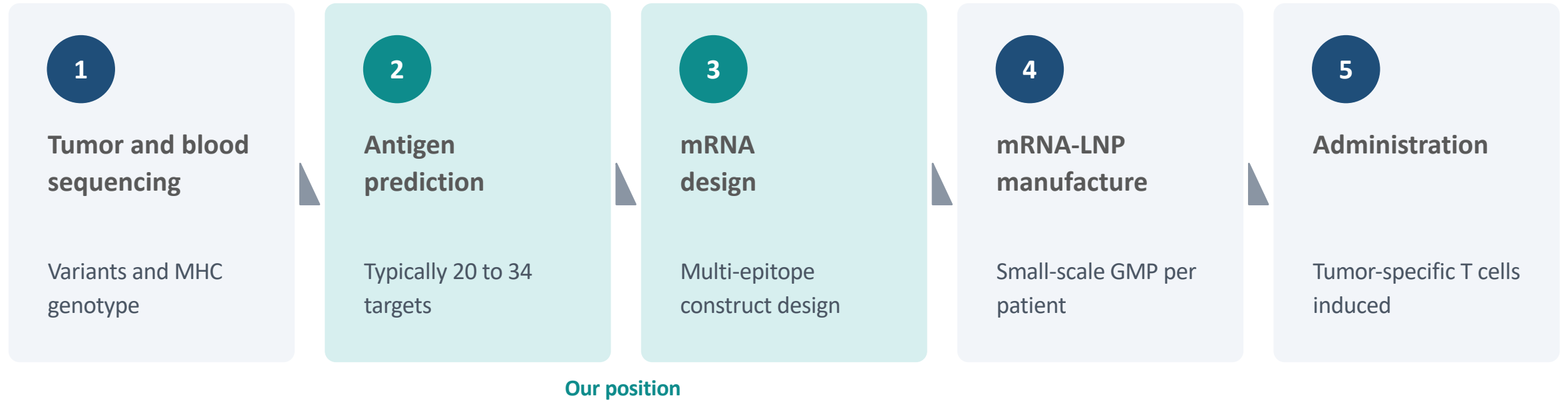
Risk reduction in recurrence and metastasis (Ph 2b)

Why this matters for us

- **The category is no longer speculative**
Regulatory filings are now in preparation
- **Expanding beyond melanoma**
Nine Phase 2/3 trials across tumor types
- **Antigen selection decides efficacy**
Manufacturing can be outsourced; prediction cannot
- **Asia remains open**
No player with local infrastructure in Korea and Japan

Where the Value Sits in the Value Chain

Five steps from a single patient's tumor to a vaccine made for that patient



The value concentrates in vaccine design

Sequencing, mRNA synthesis and LNP formulation are all commercially available.

What sets our vaccine apart is how antigens are selected and assembled, built from a decade of research.

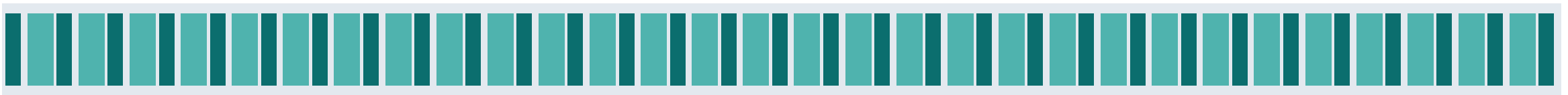
What Prediction Accuracy Changes

Accurate prediction pins down the epitope, so the construct carries targets instead of flanking insurance

Conventional design · ~27-mer × ~34 Epitope sequence 44%



Epitope-resolved design · 15-mer × 61 Epitope sequence 80%



Identical mRNA length · approximately 915 amino acids

 Class I epitope (9 aa)

 Class II and B cell epitope (15 aa)

 Flanking sequence

Flanking sequence is insurance against an uncertain epitope. Resolving the epitope frees that space for more targets.

Research Foundation

Originating in a KAIST laboratory, with peer-reviewed models forming the core of the product

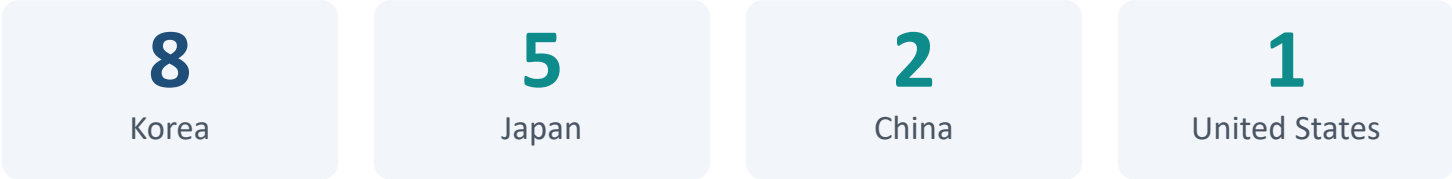


Patents granted or filed for DeepNeo-MHC, DeepNeo-TCR, DeepNeo-BCR, LexiNeo-TCR, and LexiNeo-BCR.
The platform rests entirely on proprietary intellectual property.

Intellectual Property and Publications

A patent family covering each stage of antigen prediction, grounded in peer-reviewed research

Registered patents



Key patents

Technology	Registered	Filed
MHC binding prediction	KR · JP · CN	US · EU
T cell response prediction	KR · JP · CN · US	EU
B cell epitope prediction	—	KR · PCT
Target database	KR · JP	US · EU · CN
Dependency prediction	KR · JP	US · EU · CN

Adjacent portfolio

Companion diagnostics

iMethyl® and immunotherapy response prediction

Selected publications

- DeepNeo platform**
Nat Commun 2020 · *Nat Genet* 2023 · *Nucleic Acids Res* 2023 · *Sci Adv* 2025
- Target database**
Nat Biotechnol 2023
- Dependency prediction**
Genome Biol 2020 · *Clin Transl Med* 2022
- Companion diagnostics**
Nat Commun 2019 · *Nat Cancer* 2023



MHC II immunogenicity shapes the neoepitope landscape in human tumors

Nature Genetics, 2023

An Integrated AI Platform

Beyond MHC binding, predicting whether immune cells actually recognize the antigen and respond

DeepNeo

CNN · patent granted

In service

MHC presentation and TCR/BCR response

LexiNeo

LLM · patent filed

Validation 2026-27

Immune repertoire context

DarkNeo

Foundation model

Development 2026-28

Next-generation target discovery

Core modules

MHC-I and MHC-II

Probability of cell-surface presentation

TCR immunogenicity

Probability of T cell recognition

BCR immunogenicity

B cell uptake and T cell stimulation

Repertoire language model

Cross-attention immunogenicity prediction

Output: a prioritized antigen list for each patient and a multi-epitope mRNA design

Pipeline by Antigen Class

The class of antigen determines whether the vaccine is individualized or off-the-shelf

	Tumor-Associated Antigen	Neoantigen	Cryptic Antigen
Origin	Overexpression of normal proteins	Somatic mutation	Non-canonical translation (dark proteome)
Tumor specificity	Low	Very high	Potentially high
Immunogenicity	Low	High	Potentially high
Shared across patients	Shared	Patient-private	Potentially shared
Vaccine format	Off-the-shelf or individualized	Individualized	Off-the-shelf or individualized
Our approach	DeepNeo and LexiNeo individualized	DeepNeo and LexiNeo individualized	DarkNeo discovery, population-tailored

Cryptic Antigens

Translated from the dark proteome outside standard gene annotation, tumor-specific yet shared across patients

What they are

Peptides translated from non-coding regions
Only recently shown to be produced in tumors

Opportunity in the dark proteome

Highly immunogenic, escaping central tolerance
Recurring across patients with the same tumor type

Role of DarkNeo

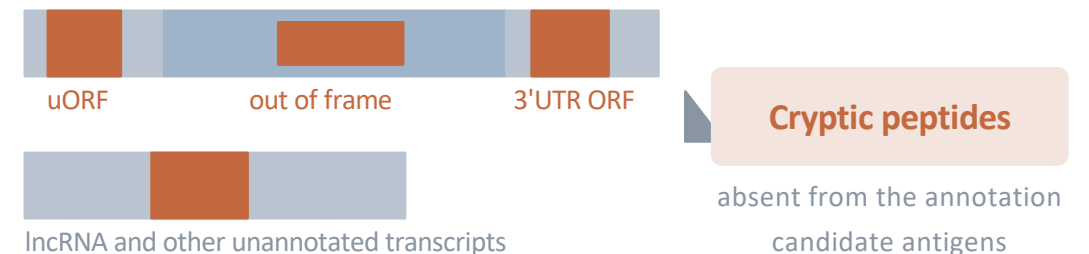
Infers translation from transcriptional patterns
Discovers tumor-specific non-canonical translation

Where cryptic antigens come from

Annotated ORF



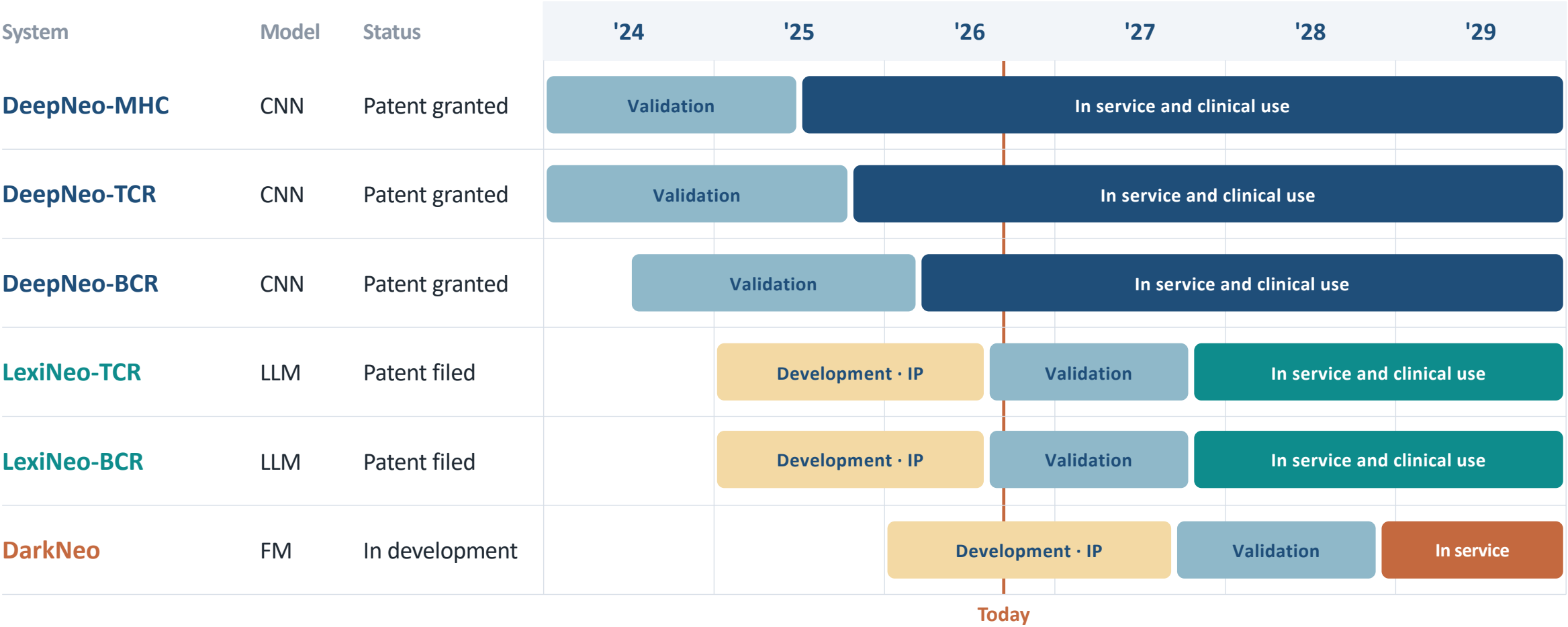
Non-canonical translation



Shared cryptic antigen panels per tumor type lead to an off-the-shelf mRNA vaccine (DLV-C1)

Roadmap: AI Systems

DeepNeo, LexiNeo and DarkNeo



DLV Vaccine Pipeline

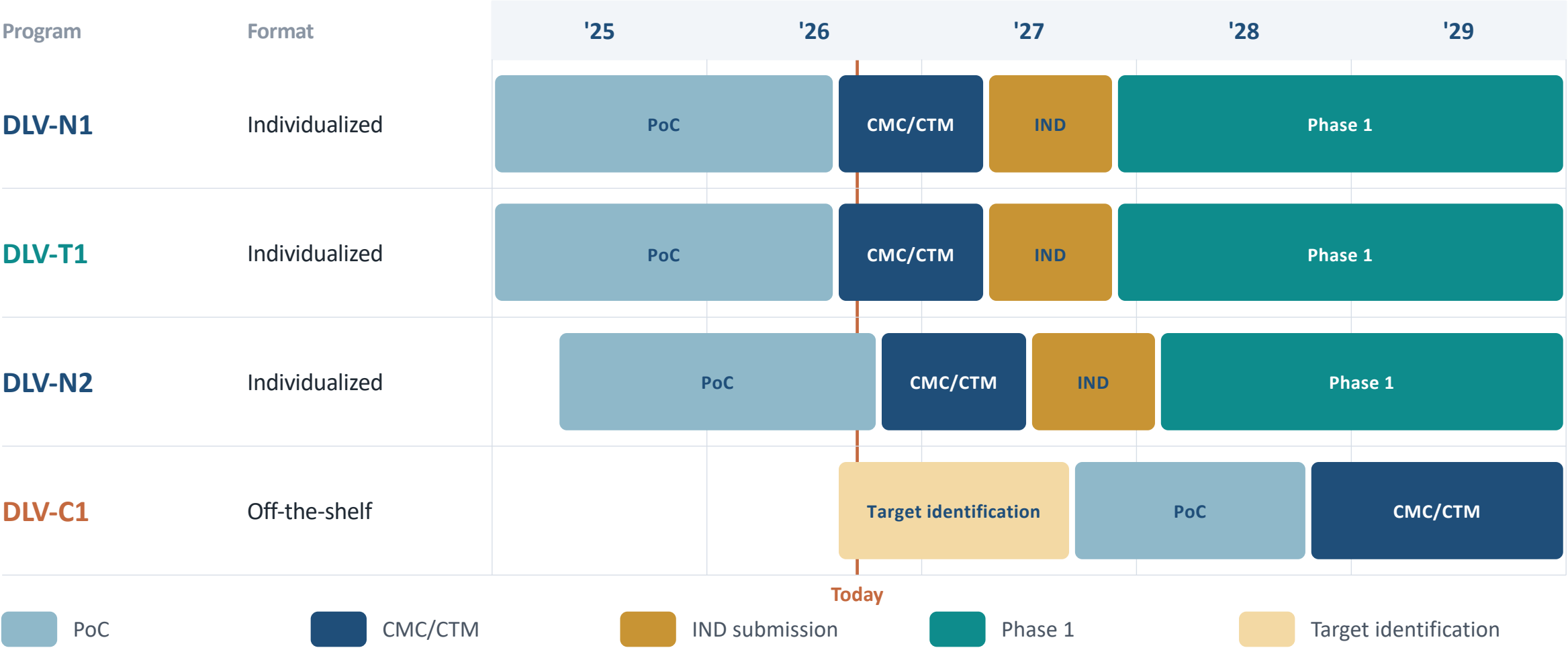
Individualized programs entering the clinic, followed by an off-the-shelf vaccine

Program	Target antigen	Format	AI system	Indication	Current stage
DLV-N1	Neoantigen	Individualized	DeepNeo + LexiNeo	Hepatocellular carcinoma Adjuvant after resection	PoC to CMC/CTM
DLV-T1	Tumor-associated antigen	Individualized	DeepNeo + LexiNeo	Ovarian cancer Maintenance	PoC to CMC/CTM
DLV-N2	Neoantigen	Individualized	DeepNeo + LexiNeo	Gastric cancer With adjuvant chemotherapy	PoC
DLV-C1	Cryptic antigen	Off-the-shelf	DarkNeo + DeepNeo + LexiNeo	Indication under selection (joint development)	Target identification

Modality for all programs: mRNA-LNP

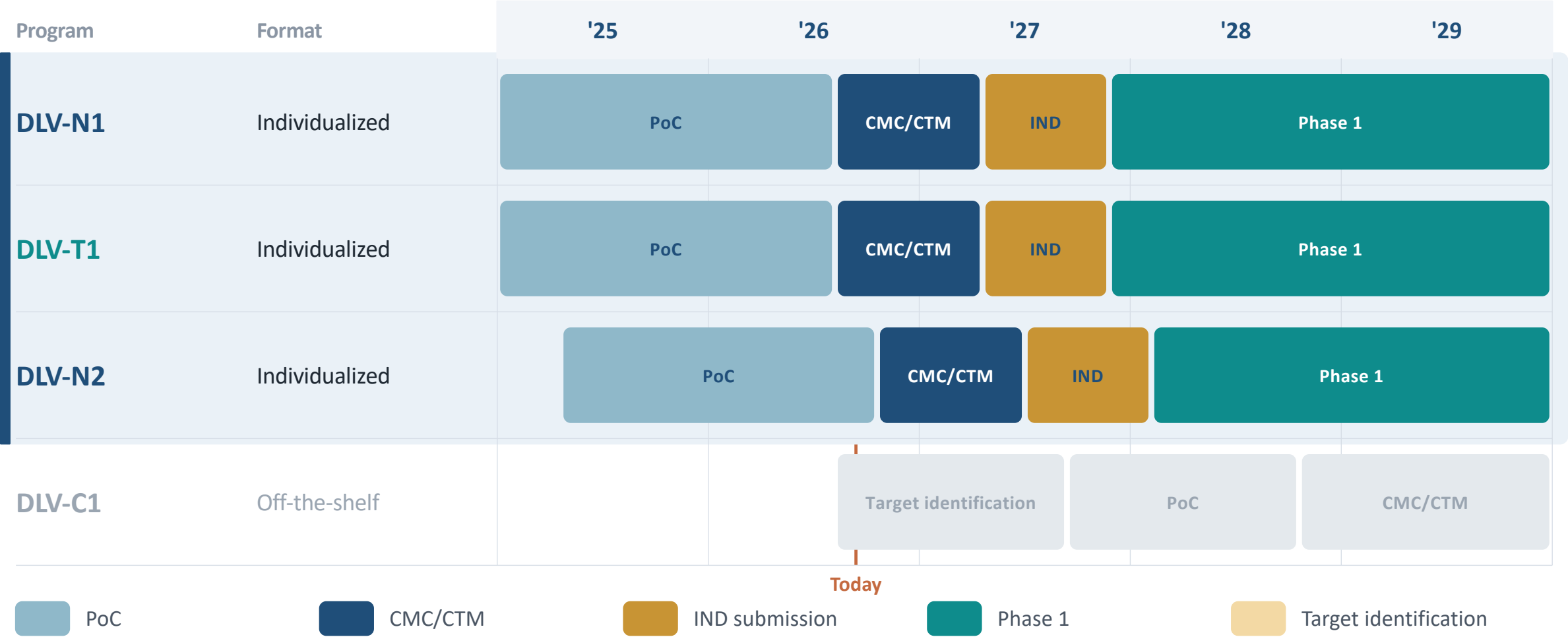
Roadmap: Vaccine Pipeline

Individualized programs reach IND in 2027; the off-the-shelf program follows



Roadmap: Individualized Programs

Three individualized programs enter the clinic first, with IND submission from 2027



Clinical Entry Strategy

Simultaneous entry in Korea and Japan, in settings of minimal residual disease where the vaccine is strongest

DLV-N1

Hepatocellular carcinoma

Adjuvant after resection

Phase 1 Vaccine

Phase 2 Surveillance vs Vaccine

or

Advanced, first or second line

Safety run-in ICI + Vaccine

Phase 2 ICI vs ICI + Vaccine

Priority 1

DLV-T1

Ovarian cancer

Maintenance after chemotherapy

Phase 1 Vaccine

Phase 1b PARPi vs PARPi + Vaccine

or

Recurrent, after PARPi

Safety run-in ICI + Vaccine

Phase 2 BSC vs ICI + Vaccine

Priority 2

DLV-N2

Gastric cancer

With adjuvant chemotherapy

Phase 1 Chemotherapy + Vaccine

Phase 2 Chemo vs Chemo + Vaccine

Priority 3

Priority determined by recurrence rate, unmet need and the strength of evidence for combination

Key Milestones to Clinical Entry

From CMC proof of concept through CDMO contract and CTM production to IND submission

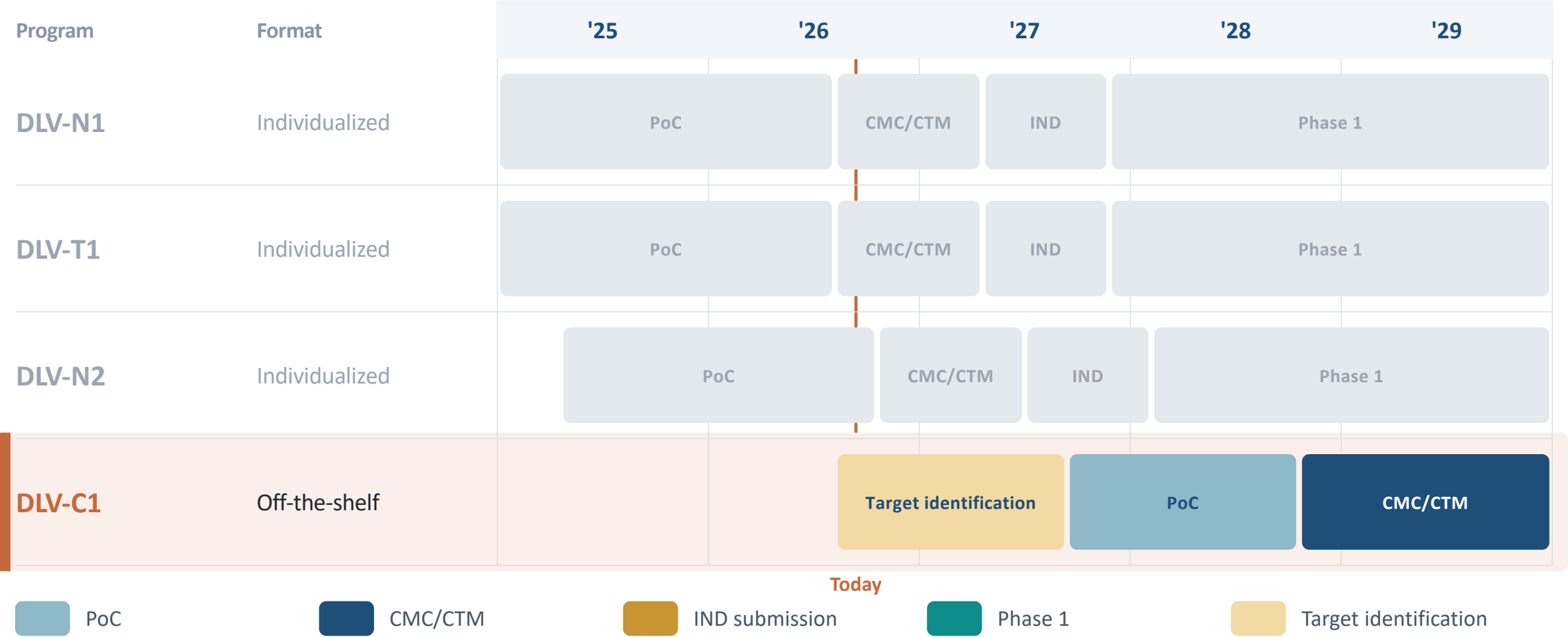


Clinical design strategy

Phase 1 is designed around DLV vaccine monotherapy to establish safety and immunogenicity, with combination regimens introduced in Phase 1b or Phase 2.

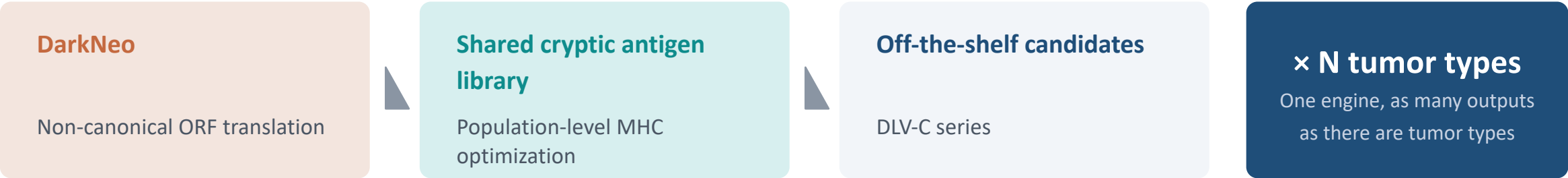
Roadmap: Off-the-shelf Program

The off-the-shelf program follows, entering development once target identification is complete



Core Discovery Engine

A discovery platform that keeps producing off-the-shelf vaccine candidates



Applicable tumor types, illustrative. Highlighted types have low mutational burden and are difficult to address with existing vaccines



A widening set of commercial opportunities



Manufacturing and Clinical Partners

Neogenlogic concentrates on vaccine design while established partners handle manufacturing and clinical operations

Manufacturing

Partner names will be disclosed in line with contract and disclosure timelines

Leading domestic CDMO

mRNA drug substance

Global oligonucleotide CDMO with FDA- and PMDA-inspected sites, and its own mRNA vaccine taken through Phase 1

Leading LNP manufacturer

Delivery technology

Its own mRNA-LNP platform, free to operate, now in Phase 2 with a COVID-19 vaccine and extending into oncology

Leading vaccine manufacturer

Joint research · IND development

Korea's leading vaccine maker, with integrated GMP manufacturing and an established commercial network

Companies holding ICI or PARPi

Partnership target

Global oncology companies, for combination trials and global co-development

Clinical

Ovarian

Severance Hospital

DLV-T1 · maintenance with PARPi or ICI

Hepatocellular

Bundang CHA Hospital

DLV-N1 · adjuvant with ICI

Gastric

Site being confirmed

DLV-N2 · with CAPOX or S-1

CRO in Japan

Leading Japanese CRO

Clinical operations · PMDA
Local manufacturer sourcing

Why Neogenlogic

Neogenlogic aims to lead the development of AI-based cancer vaccines in Asia

1 Validated science, validated modality

Phase 3 proof for the category

Peer-reviewed models behind our prediction

2 Defensible position in the value chain

Manufacturing is procured

Prediction is proprietary and patented

3 Clinical entry in 2027

Three programs, Korea and Japan

First patient dosed targeted for 2028

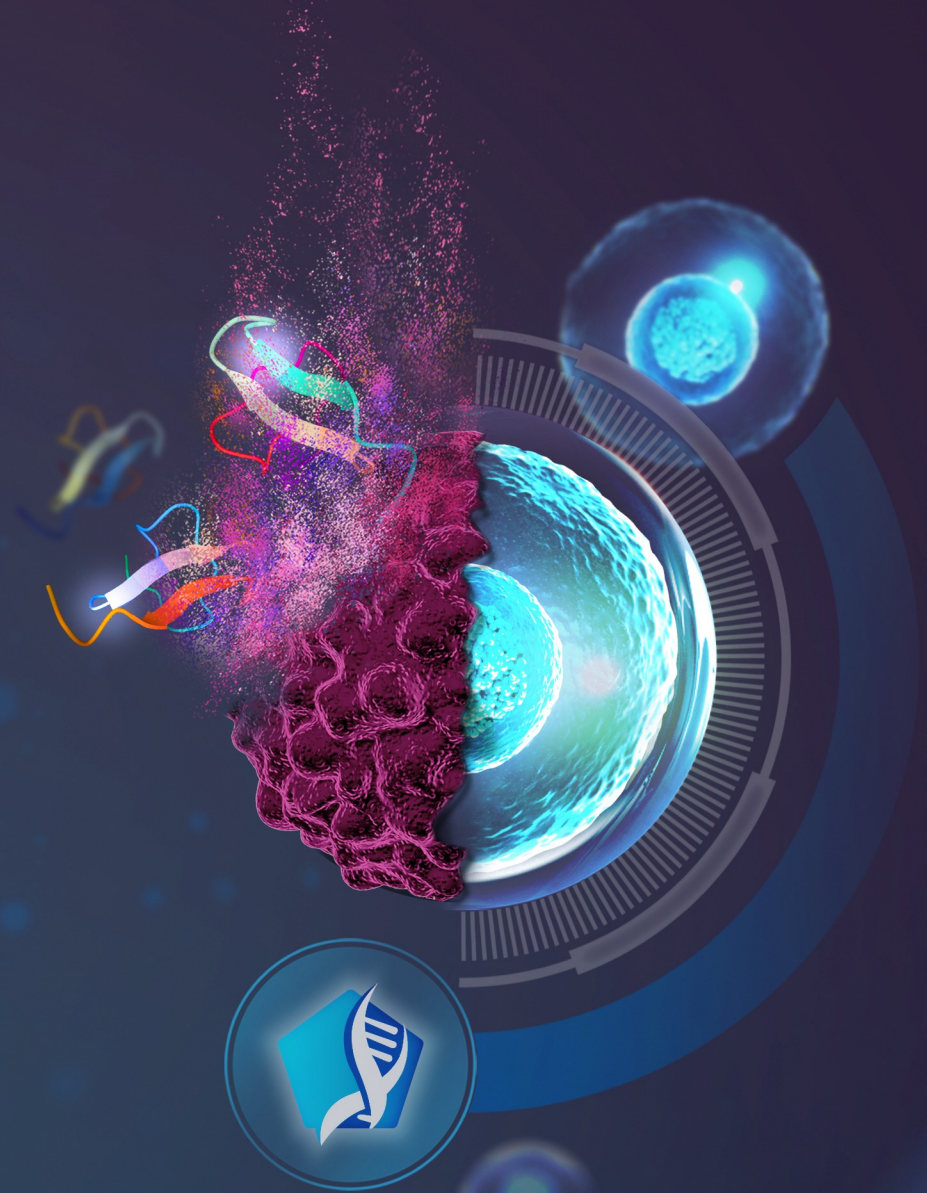
4 A second engine beyond individualization

Off-the-shelf candidates, tumor by tumor

Licensing and partnership revenue

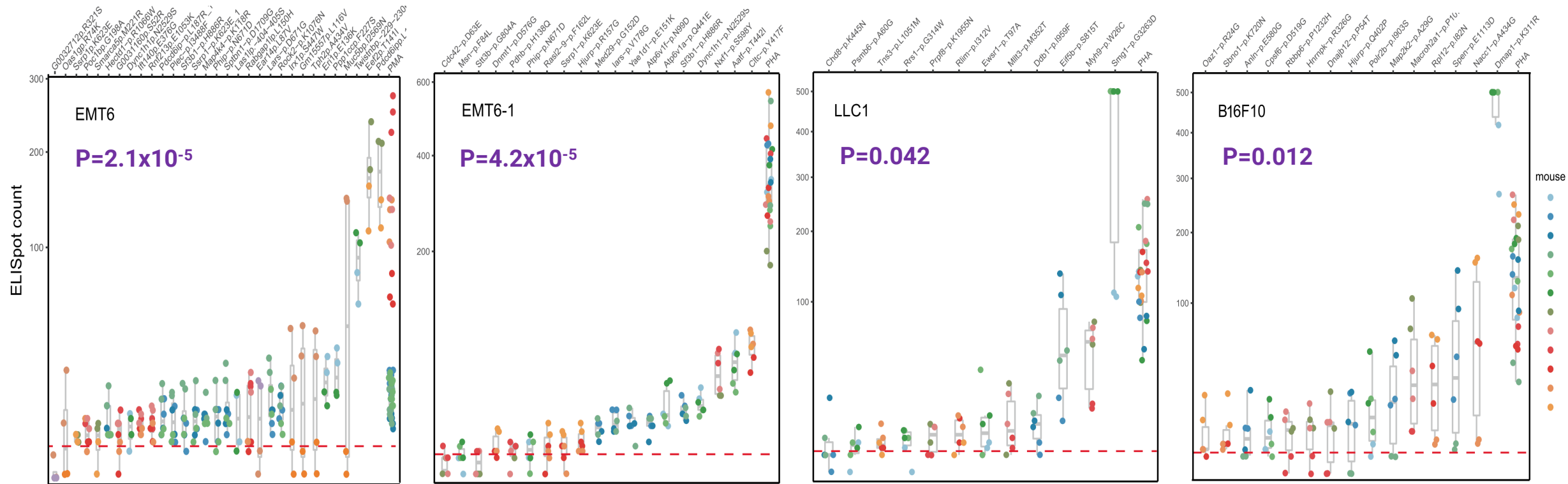
*Empowering AI
to Bring the Right Vaccine to Your Cancer*

Neogenlogic



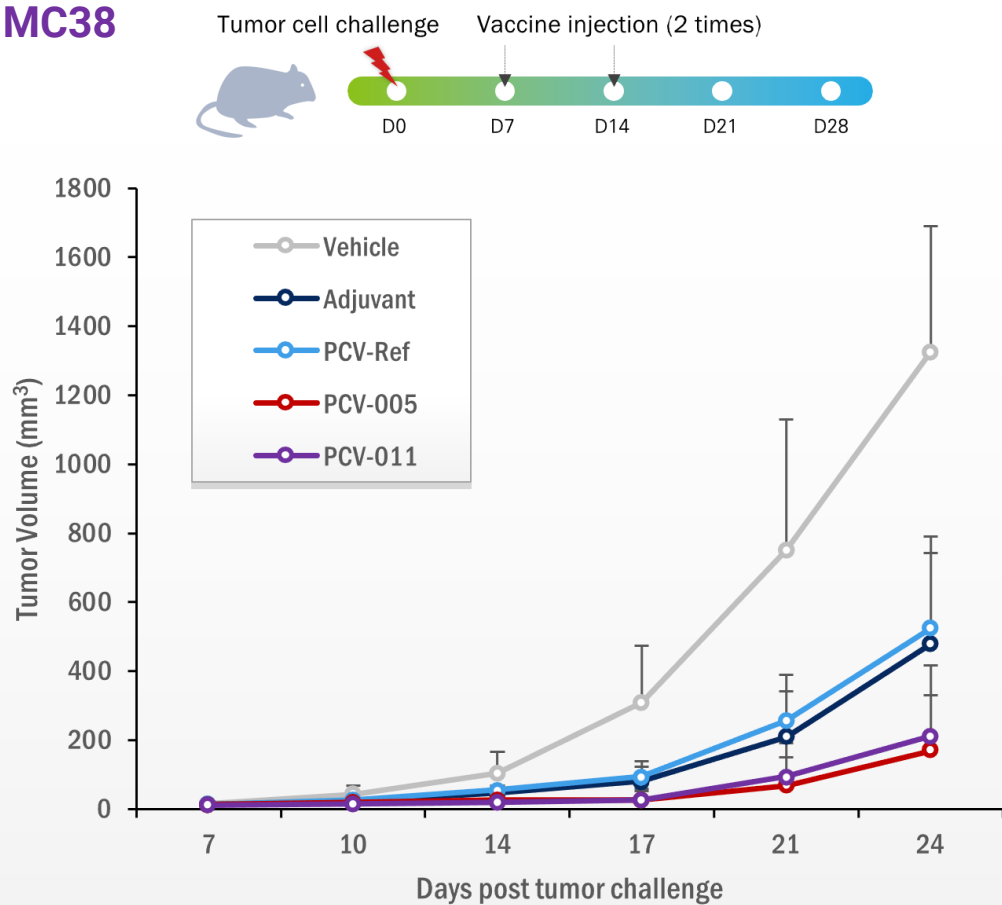
Vaccination followed by ELISpot assay of splenocytes in mice

75 vaccines in 51 mice with 464 experiments

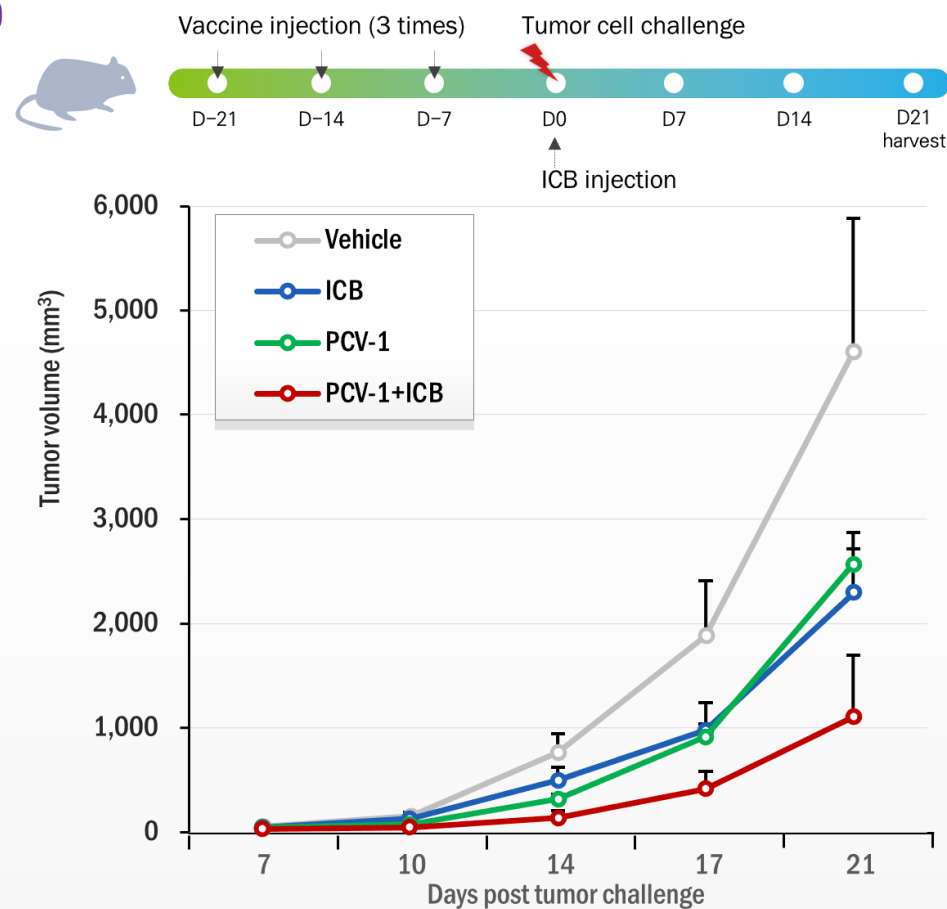


Vaccination followed by tumor regression analysis in mice

MC38

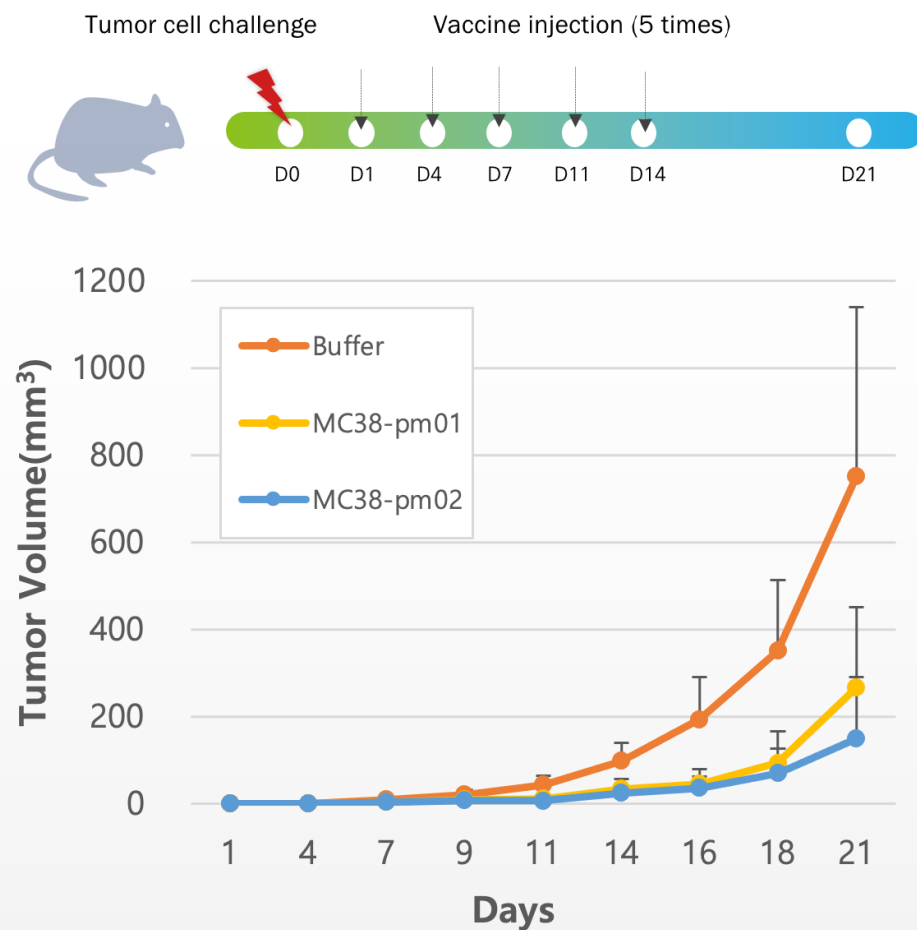


B16F10



Vaccination followed by tumor regression analysis in mice

MC38



mRNA Vaccine Construct

MC38_pm01: 10 NeoAgs

450aa, 1353nt

MC38_pm02: 20 NeoAgs

Signal peptide

Linker1

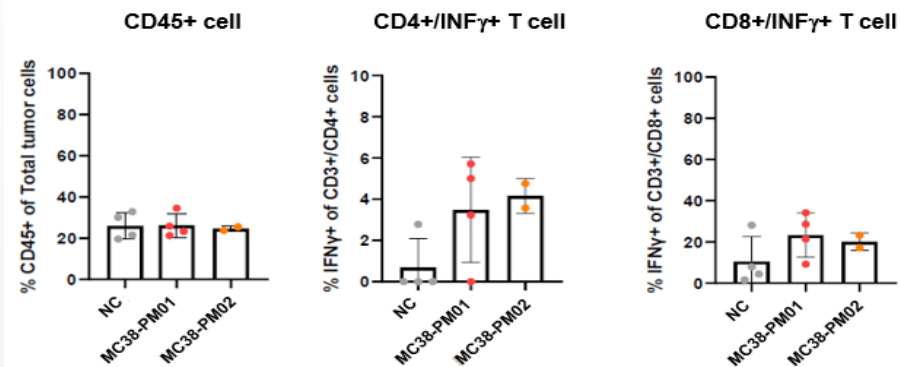
Linker2

MITD

NeoAg

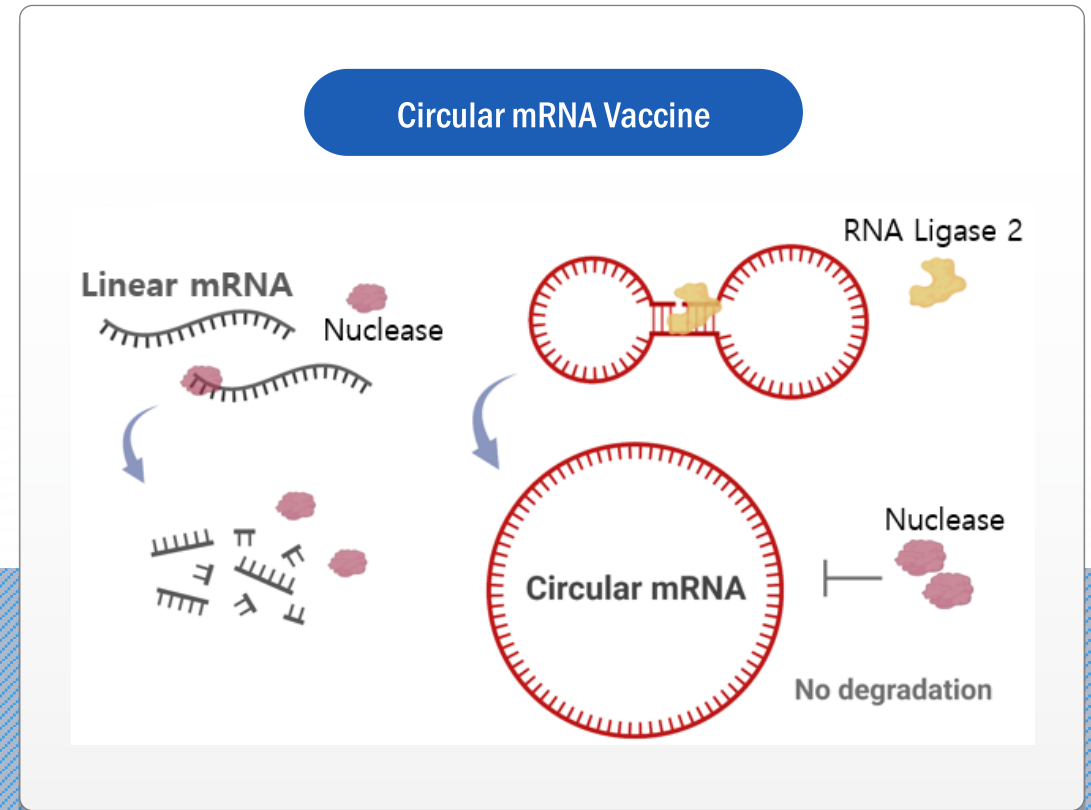
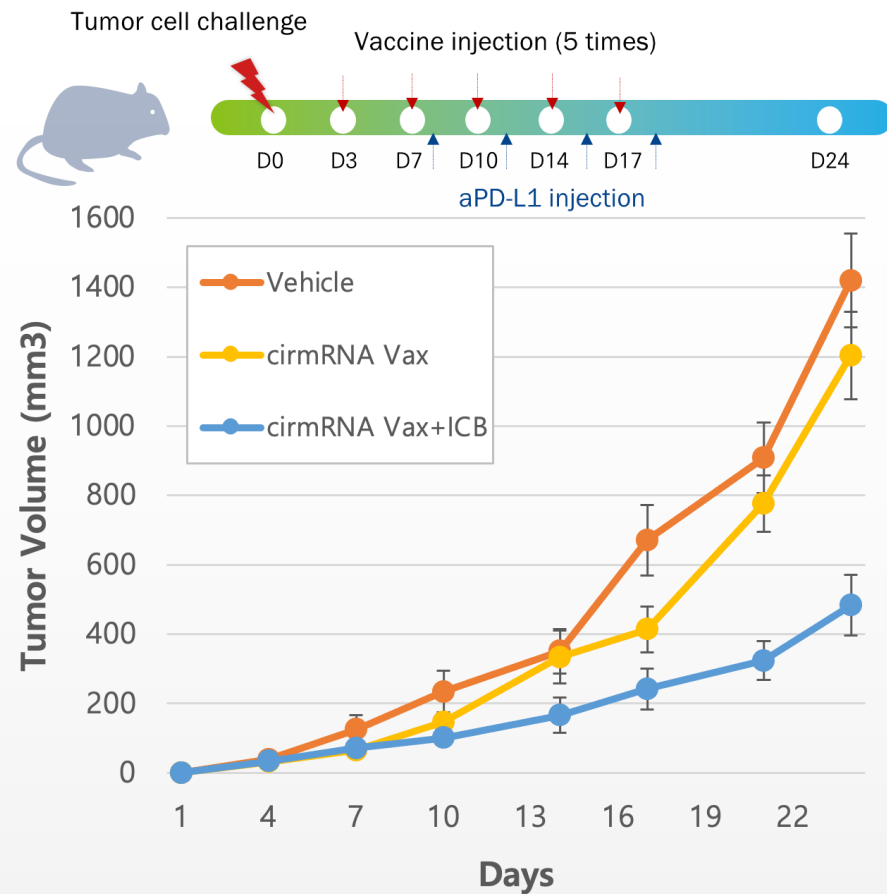
584aa, 1755nt

Immune Cells Analysis



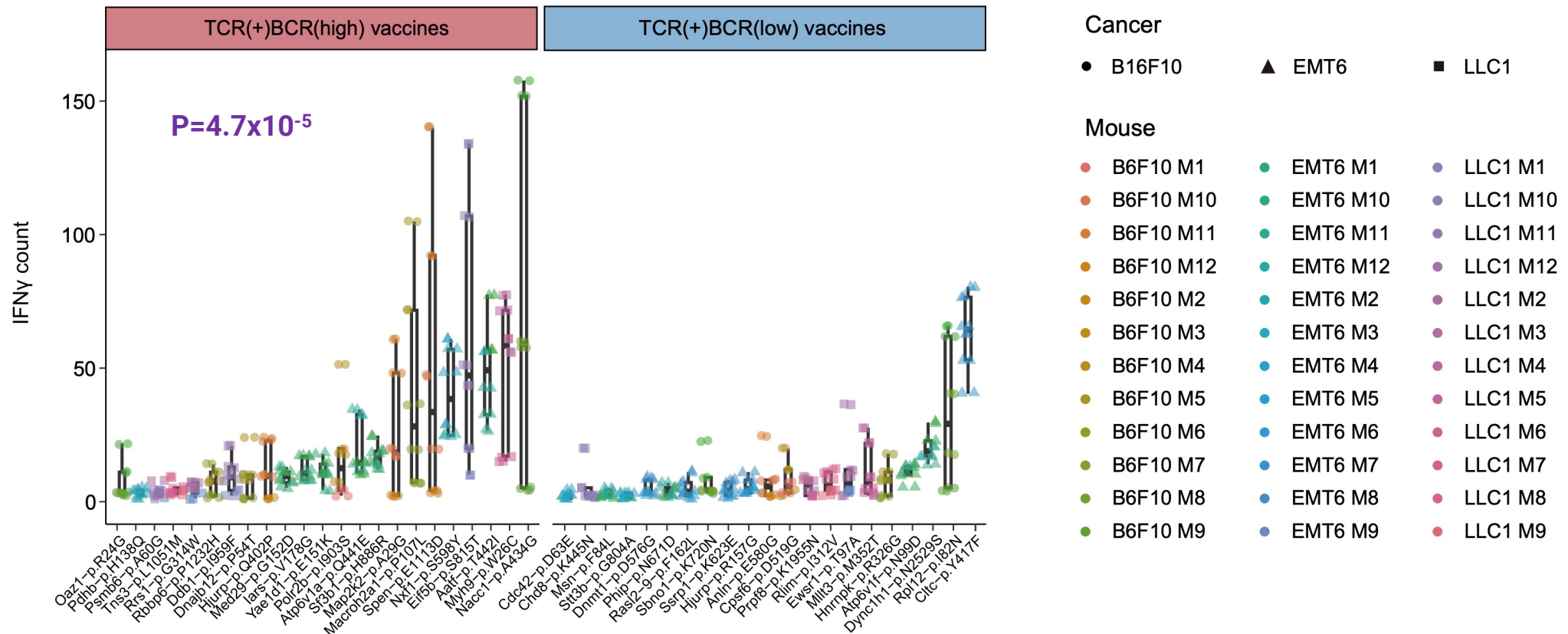
Vaccination followed by tumor regression analysis in mice

4T1



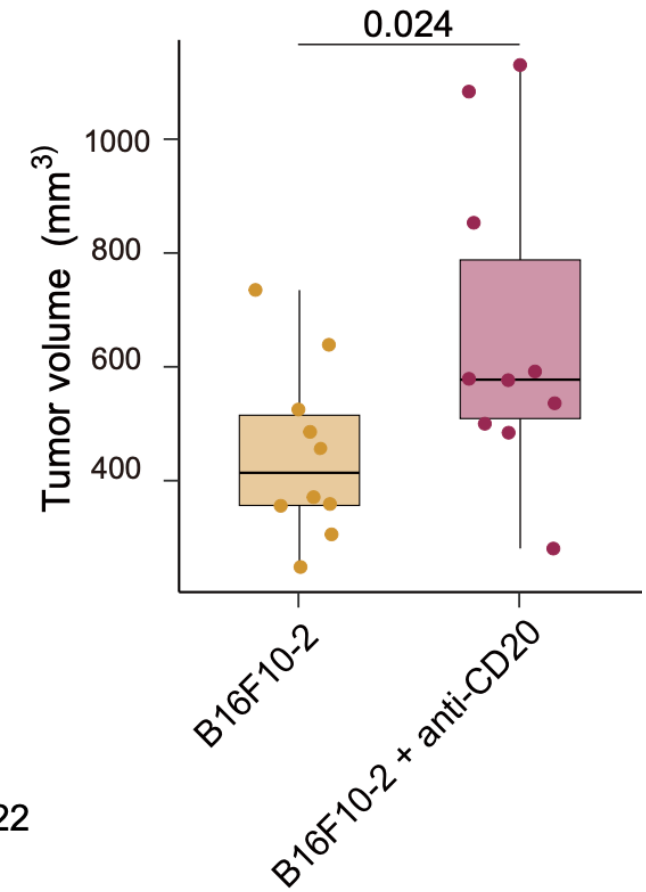
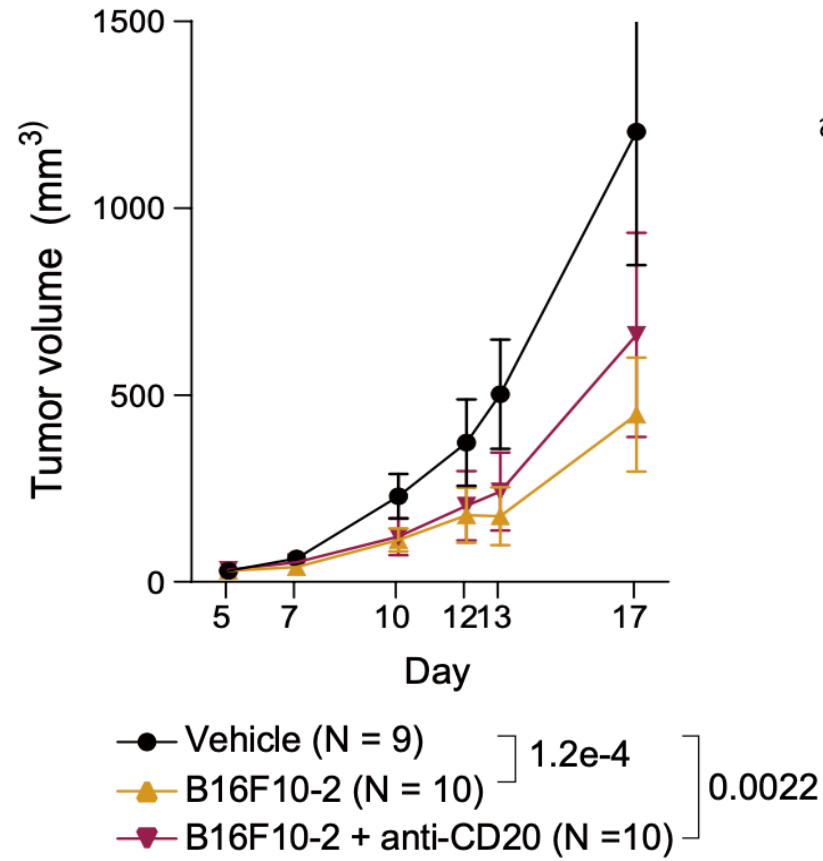
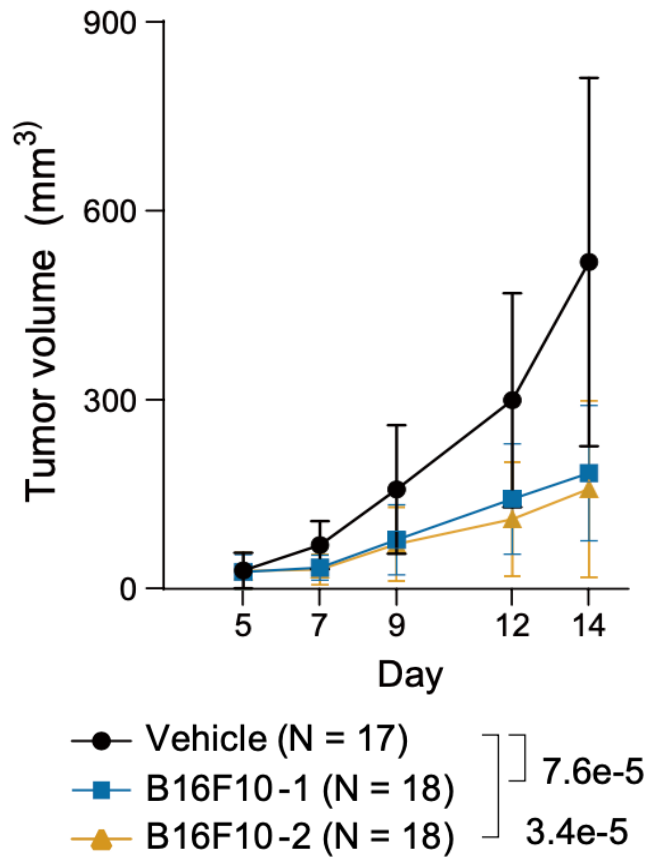
Vaccination followed by ELISpot assay of splenocytes in mice

75 vaccines in 51 mice with 464 experiments



Vaccination followed by tumor regression analysis in mice

5 vaccines for each group



Clinical Trials for Neoantigen Vaccines

nature
2017

An immunogenic personal neoantigen vaccine for patients with melanoma

SKCM



nature
2019

Neoantigen vaccine generates intratumoral T cell responses in phase Ib glioblastoma trial

GBM



Cell 2020

A Phase Ib Trial of Personalized Neoantigen Therapy Plus Anti-PD-1 in Patients with Advanced Melanoma, Non-small Cell Lung Cancer, or Bladder Cancer

BLCA



JCI

Cancer

2020

na
2020

nature

2024

penicillamine in advanced hepatocellular carcinoma: a phase 1/2 trial

HCC



CANCER DISCOVERY

2024

T-cell Responses to Individualized Neoantigen Therapy mRNA-4157 (V940) Alone or in Combination with Pembrolizumab in the Phase 1 KEYNOTE-603 Study

SKCM
NSCLC



Cell 2025

A multi-adjuvant personal neoantigen vaccine generates potent immunity in melanoma

SKCM



nature
2025

RNA neoantigen vaccines prime long-lived CD8⁺ T cells in pancreatic cancer

PDAC



nature
2025

A neoantigen vaccine generates antitumour immunity in renal cell carcinoma

RCC

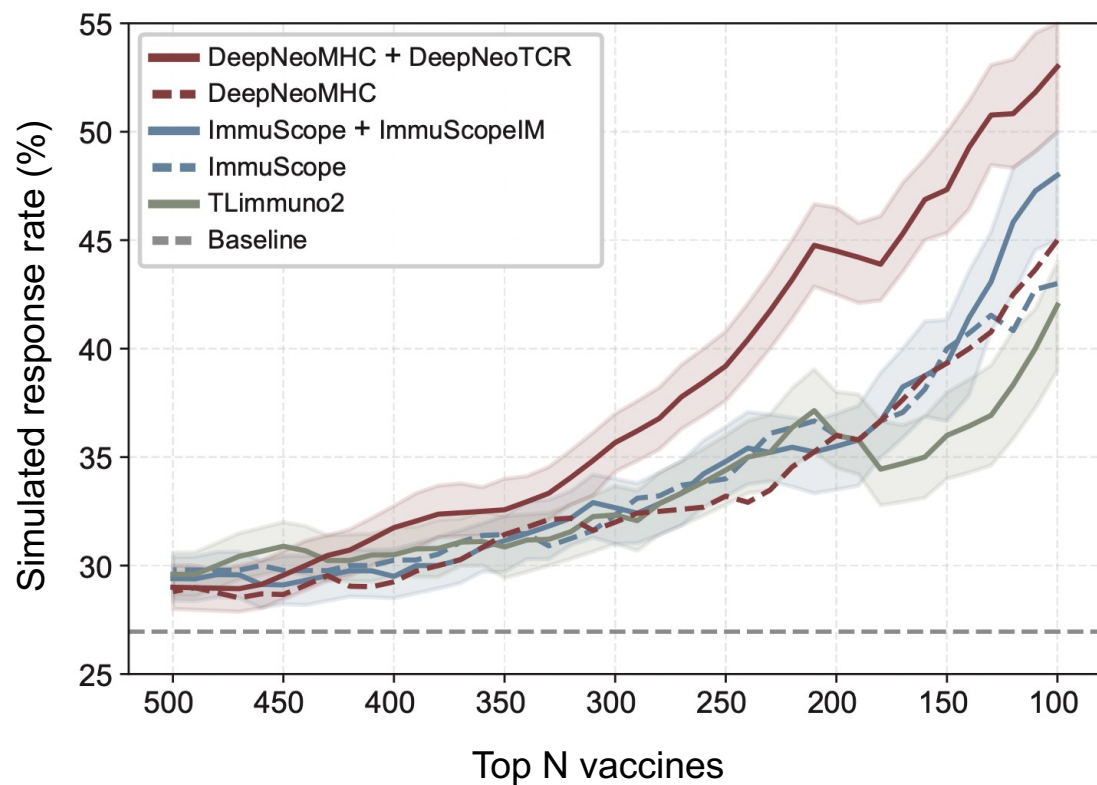


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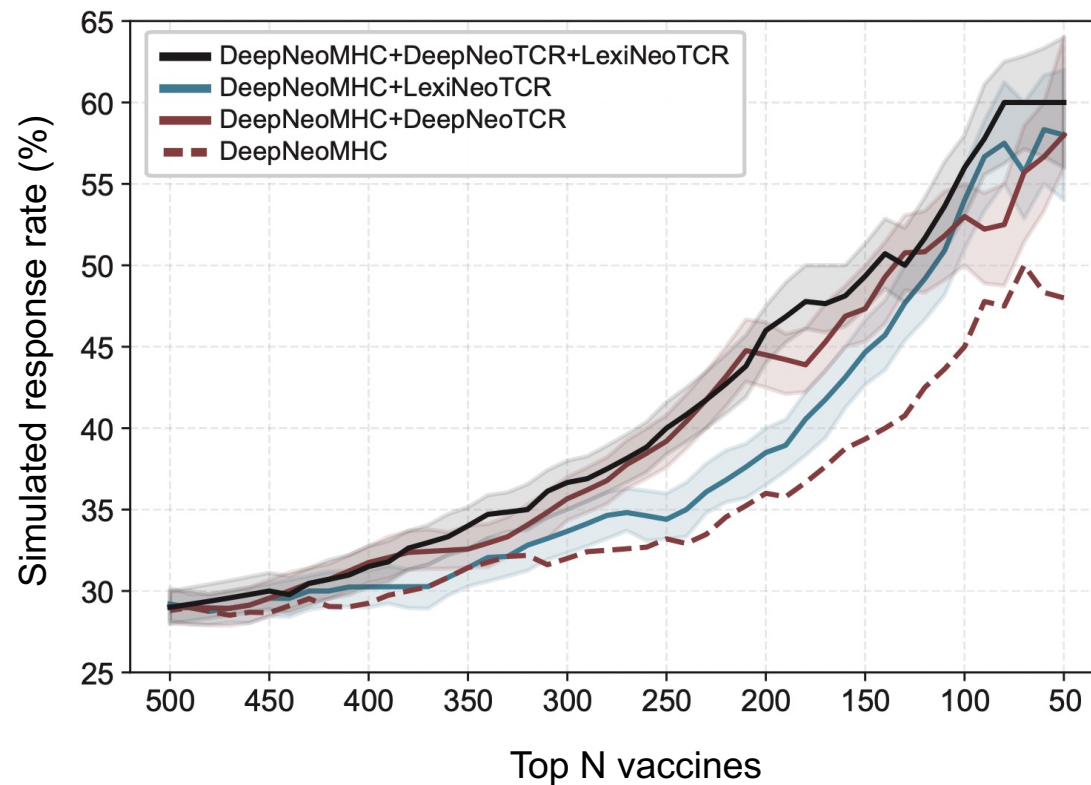
mRNA or peptide vaccines

Retrospective analysis of clinical trial results

Effects of DeepNeo-TCR



Effects of LexiNeo-TCR



Retrospective analysis of clinical trial results

Effects of DeepNeo-BCR and LexiNeo-BCR

